

scripts/check-no-guessing-vocabulary.sh — User Guide

Last verified: 2026-08-26 (§6.J corpus floor — an unresolved or empty scan-path set now refuses instead of reporting "gate clean"; LVA vacuous-pass sweep F14) **Inheritance:** HelixConstitution §11.4.6 (no-guessing-vocabulary mandate) + Lava §6.AD.6 (extracted gate)

Overview

Standalone gate that scans tracked status / closure / forensic-anchor files for forbidden guessing vocabulary. Extracted from scripts/check-constitution.sh's embedded block on 2026-05-17 so the gate is independently testable.

Forbidden words: likely, probably, maybe, might, possibly, presumably, seems to, appears to, seemingly, apparently, perhaps, supposedly, conjectured.

When a forbidden word appears in a scanned file, the gate fires unless one of these whitelist conditions matches the SAME LINE:

1. The line is prefixed with UNCONFIRMED: / UNKNOWN: / PENDING_FORENSICS: (any case).
2. The line contains a verbatim-quote marker: forensic anchor / verbatim operator|agent|user / historical quote.

These exemptions exist so:

- Hypothesis lines explicitly marked as PENDING evidence pass (tracked-task discipline).
- Forensic anchors that quote prior operator/agent output verbatim aren't flagged for the historical content's vocabulary.

Why this gate exists

Per HelixConstitution §11.4.6: "Either prove the cause with captured evidence and state it as fact, OR explicitly mark UNCONFIRMED: / UNKNOWN: / PENDING_FORENSICS: with a tracked-task ID."

The gate enforces this discipline at pre-push time so closure logs and forensic anchors don't drift into hedged language that would obscure what's actually known vs. assumed.

Usage

Default — scan production paths

```
bash scripts/check-no-guessing-vocabulary.sh
```

Scans:

- `.lava-ci-evidence/sixth-law-incidents/`
- `.lava-ci-evidence/crashlytics-resolved/`

Across `*.md` + `*.json` files.

Custom scan paths (hermetic-test mode)

```
LAVA_NO_GUESSING_SCAN_PATHS="path1:path2:path3" bash scripts/  
check-no-guessing-vocabulary.sh
```

Colon-separated list. Used by `tests/check-constitution/test_no_guessing_vocabulary.sh` to scan fixture dirs without touching `.lava-ci-evidence/`.

Custom repo root (hermetic-test mode)

```
LAVA_REPO_ROOT="/path/to/synthetic/repo" bash scripts/check-no-  
guessing-vocabulary.sh
```

Exit codes

Code	Meaning
0	Gate clean — no forbidden vocabulary found, and at least one file was actually read. The verdict line now names the corpus: §11.4.6 no-guessing vocabulary gate clean: <N> file(s) scanned across <M> director(ies).
1	At least one violation (paths printed to stderr), or the scan corpus was empty / partial (added 2026-08-26 — see below)

2026-08-26 update — §6.J corpus floor: "gate clean" now requires having read something

The defect

The scan loop opened with `[[ -d "$p" ]] || continue`, which skipped a missing scan path **in silence**. A configuration in which no path resolved therefore produced:

```
LAVA_NO_GUESSING_SCAN_PATHS=<nonexistent> -> "$11.4.6 no-guessing vocabulary gate clean." exit 0
```

and so did a path that resolved but held no `*.md / *.json`. Both are "nothing was learned" reported as "nothing failed" — the shape §6.J forbids, and the same shape `scripts/check-constitution.sh`'s clause-6.H credential floor already guards against.

The silent-skip also made the gate's verdict a function of **checkout state**: a tree where `.lava-ci-evidence/crashlytics-resolved/` happened not to exist yet scanned one directory instead of two and said nothing about the difference.

What the gate now refuses

The loop counts what it resolves and what it reads, and three floors run **before** any clean verdict can be printed. All exit **1**:

Condition	Message
No configured scan path resolved to a directory	VIOLATION: NO scan path resolved to a directory. — every configured path is listed with (MISSING)
Paths resolved but zero <code>*.md / *.json</code> files were read	VIOLATION: the scan read ZERO files. — resolved directories are listed as (present, 0 matching files), unresolved ones separately
Some paths resolved and some did not	VIOLATION: the scan examined a PARTIAL corpus. — the missing paths are named

The first floor distinguishes its cause from the environment: if the paths came from `LAVA_NO_GUESSING_SCAN_PATHS` this is a **configuration error**, and the remedy is to correct that variable (colon-separated, relative to the repo root) or unset it to fall back to the built-in defaults; if they are the **built-in defaults**, then either this is not a Lava checkout or `LAVA_REPO_ROOT` points elsewhere — and the message prints the resolved root so the reader can see which.

The partial-corpus floor exists because a floor that fires only at exactly zero is a floor with one stair. Its remedy line offers both honest options: restore the missing path, **or** narrow LAVA_NO_GUESSING_SCAN_PATHS to exactly the set you intend to audit — so the gate's claim matches the corpus it actually read.

Effect on the hermetic test

tests/check-constitution/test_no_guessing_vocabulary.sh drives this gate through LAVA_NO_GUESSING_SCAN_PATHS against fixture directories. Fixtures must now point at directories that exist and contain at least one *.md or *.json — a fixture path that was quietly skipped before will now fail the gate, which is the point.

§6.J falsifiability rehearsal of the floor itself

1. LAVA_NO_GUESSING_SCAN_PATHS=/nonexistent/a:/nonexistent/b bash scripts/check-no-guessing-vocabulary.sh → expect **exit 1**, NO scan path resolved, both paths listed (MISSING), and the configuration-error branch of the diagnosis. Before this change: exit 0, gate clean.
2. mkdir -p /tmp/empty-scan && LAVA_NO_GUESSING_SCAN_PATHS=/tmp/empty-scan ... → expect **exit 1**, the scan read ZERO files.
3. Run with no override in the Lava tree → expect exit 0 and a verdict line naming the file and directory counts actually scanned.

Integration

scripts/check-constitution.sh (line ~575) delegates to this script. The pre-push hook invokes check-constitution.sh which transitively runs this gate.

§6.J anti-bluff falsifiability rehearsal

1. Add a tracked file under .lava-ci-evidence/sixth-law-incidents/ with a line like: The root cause likely involves a race condition.
2. Run bash scripts/check-no-guessing-vocabulary.sh → expect exit 1 + the file listed in stderr.
3. Prepend UNCONFIRMED: to the line. Re-run → expect exit 0.
4. Remove the test file.

The deliberate-break rehearsal proves the gate fires on real violations and that the whitelist works.

Hermetic test

tests/check-constitution/test_no_guessing_vocabulary.sh — 7
fixtures covering:

1. Clean fixture passes
2. Forbidden word without whitelist → reject
3. UNCONFIRMED: whitelist passes
4. PENDING_FORENSICS: whitelist passes
5. Verbatim-quote exemption (line-scoped) passes
6. Multiple scan paths via colon separator
7. Real-repo sanity check

Run the test:

```
bash tests/check-constitution/test_no_guessing_vocabulary.sh
```

Why intentionally NOT scanned

The gate intentionally does NOT scan CLAUDE.md / AGENTS.md / CONSTITUTION.md because those documents must DESCRIBE the forbidden vocabulary as part of the mandate itself. The gate exists for future status reports / closure logs / commit-template files, not the rule's own text.

Classification: project-specific (gate's content list is universal per HelixConstitution §11.4.6 but scan-path defaults are Lava-specific).